MOONES BERTINA

MSc Student, Transportation Planning | Tarbiat Modares University, Tehran, Iran

Email | Academic email | LinkedIn | GitHub | Website

Summary

MSc student in Transportation Planning working on multi-agent reinforcement learning (MARL) for coordinated traffic signal control. Background in traffic analysis and travel demand modeling (SUMO, PTV Visum). Seeking a PhD position starting Fall 2027 in intelligent transportation systems and RL-based traffic control.

Research Interests

Adaptive Traffic Signal Control | Multi-Agent Reinforcement Learning | Intelligent Transportation Systems | Network Demand Modeling

Education

MSc, Transportation Planning (GPA: 17.8/20) Sep. 2025 – Present (expected Sep. 2027)
Tarbiat Modares University Tehran, Iran

- Thesis: multi-agent reinforcement learning for coordinated traffic signal control along an urban corridor. Supervisor: Supervisor: Prof. Mahmoud Saffarzadeh.
- Relevant courses: Machine Learning (ongoing semester), Transportation Demand Analysis (17 out of 20), Transport Planning (18 out of 20), System Analysis (17 out of 20), Advanced Traffic Engineering (17.5 out of 20)

BSc, Civil Engineering (GPA: 3.1/4) Sep. 2019 – Sep. 2023
Isfahan University of Technology Isfahan, Iran

Research Experience

Research Assistant (part-time), Transportation Lab 2026 – Present
Tarbiat Modares University Tehran, Iran

- Designing a MARL-based coordinated signal control study in SUMO (TraCI, Stable-Baselines3, PettingZoo), to be compared against classical fixed-time coordination baselines.
- Developing SUMO-based demand and traffic assignment models for Tehran land uses (hospitals, university campus, recreational center) using the Tehran Trip Generation Manual and a TAZ-level OD matrix; Python scripts generate network and zone reports.
- Team project on a traffic assignment model that reduces through-traffic in residential neighborhoods: built the OSM-based network model and estimated demand from the Tehran Comprehensive Plan.

Selected Projects

Kerman Municipality Traffic Study, Report 1: Shora Square 2026
Tarbiat Modares University (team project) Kerman, Iran

- Team study covering macroscopic modeling in PTV Visum, microsimulation of road-closure scenarios, and multi-criteria evaluation of the options. My role: report writing and analysis of traffic scenarios.

Transportation Demand Modeling, PTV Visum 2026
MSc coursework GitHub link

- Built and calibrated a 5-zone network model for Tehran: OD matrix calibration with a gravity model (KALIBRI), logit modal split, and PrT/PuT assignment. Wrote the full analytical report and published the project on GitHub.

Systematic Review: Deep RL (PPO) for Adaptive Traffic Signal Control June 2026
MSc seminar project

- Reviewed the literature on deep reinforcement learning (PPO-based) methods for adaptive traffic signal control at single intersections.
- Developed a reinforcement learning signal controller for a single intersection in the SUMO simulator.Algorithms: PPO, DQN
- This review and the single-intersection controller are the foundation for current MSc research on network-level, multi-agent signal coordination.

Professional Experience

Traffic Analyst (full-time) Mar. 2024 – Feb. 2025
EttehadRah, transportation planning and traffic consulting Tehran, Iran

- Collected, processed, and analyzed traffic data for planning and traffic-study deliverables.
- Prepared GIS-based network inputs for demand modeling and simulation tools such as Visum.
- Evaluated intersection and corridor performance and supported recommendations for network improvements.

Structural Engineer and Designer (full-time) Dec. 2023 – Feb. 2025
Atishar, construction and architecture consulting Tehran, Iran

- Structural analysis and design (ETABS, SAP/SAFE, AutoCAD) and a Python tool that automated cost estimation.

Service and Teaching

- Executive Team Member, *Challenges in the Development and Maintenance of Airport Infrastructure* conference Dec. 2025
- Executive Team Member, *20th International Conference on Traffic and Transportation Engineering* Jan.–Feb. 2026
- Instructor (part-time), Pardisan Institute 2022 – Present

Skills

- **Programming:** Python, R, C/C++
- **ML / RL:** PyTorch, TensorFlow, Stable-Baselines3, PettingZoo, NumPy, Pandas
- **Transportation software:** SUMO (netconvert, TraCI, od2trips), PTV Visum, VISSIM
- **Spatial data:** QGIS, OpenStreetMap
- **Other:** Git, LaTeX, MS Office; ETABS, AutoCAD, Civil 3D

Standardized Tests

- IELTS Academic (Feb. 2023): 7.0 overall (Listening 8, Speaking 7, Reading 6, Writing 6); retake scheduled
- GRE General (Oct. 2024): 303 total, Quantitative 163, Verbal 140, Analytical Writing 3.0

References

Available upon request